Lecture 9 Notes

Quantitative Data Analysis

Introduction

- Today we will continue our discussion of the assumptions of the linear model
- We began last time by discussing variable selection
- That is a crucial step which begins at the start of a project
- Some of the other assumptions we meet are more technical in nature
- We begin by recapping the five assumptions

Assumptions

We will consider the following assumptions (taken from Peter Kennedy's "A guide to econometrics"):

1. The model is correctly specified
2. The residuals have the same variance and are independent of one another
3. There are no exact linear relationships between independent variables
4. The independent variables are fixed and measured without error
5. The expected value of the residual is zero

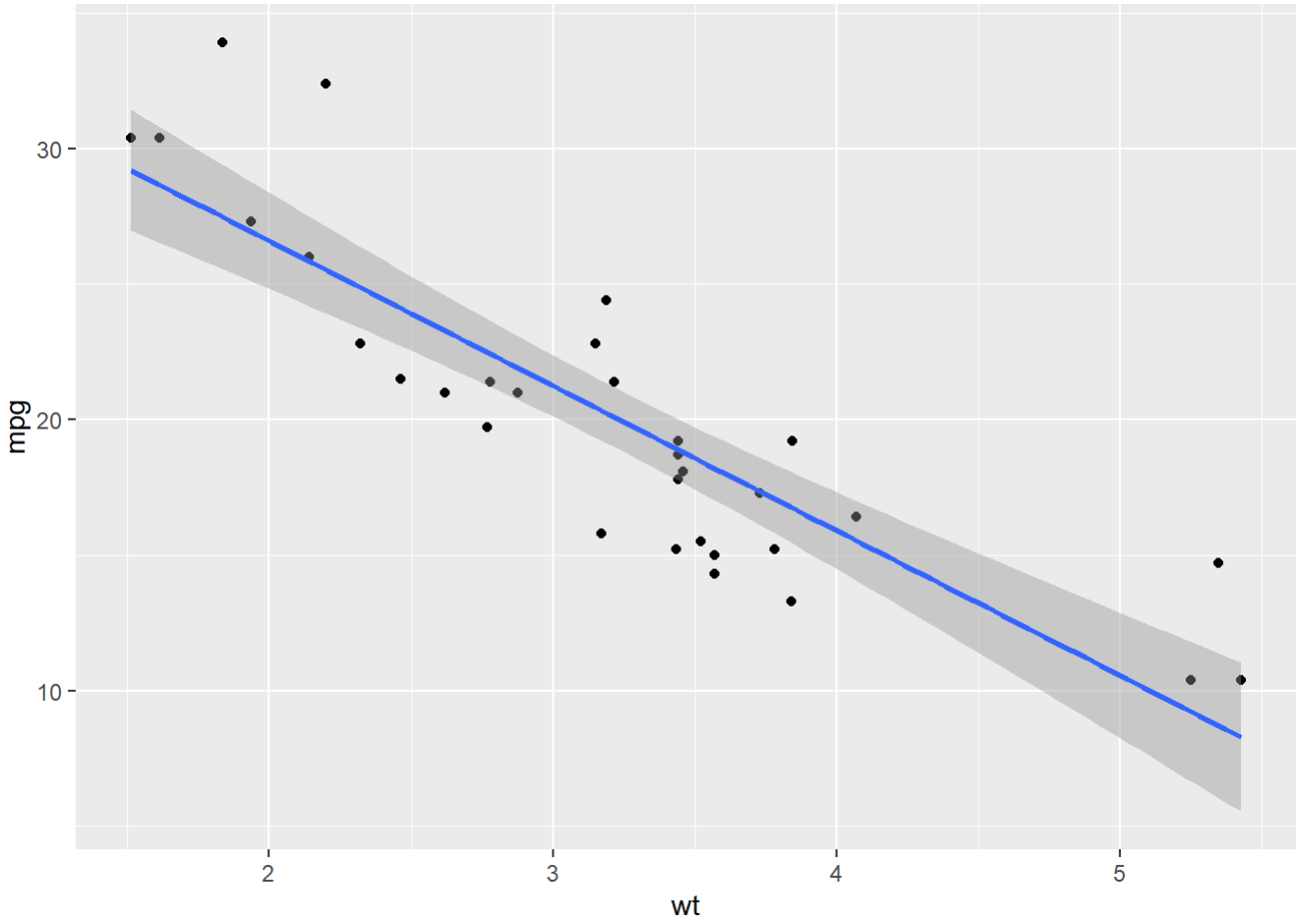
The model is well specified

- We have so far assumed a particular functional form of the relationship we are modelling
- For example, for age we assume that being one year older is associated with a fixed change in the estimated wage
- The amount does not depend on the person's wage or their age, it is a fixed sum of money which is added based on a change in age
- This doesn't sound particularly sensible for our case
- We need not constrain ourselves in this way

Visualising non-linearity

- I'm going to switch to another dataset at the moment to demonstrate this point
- R has a built in dataset called *mtcars* which contains some information about different cars
- We are going to look at the relationship between fuel efficiency (mpg) and weight (wt)

- We can load the dataset by typing data(mtcars) into R's console



Thoughts

- The relationship looks relatively linear
- However, the regression underestimates the efficiency of the lightest and heaviest cars
- It does reasonably well in the middle of the range
- One trick to dealing with this is to take a logarithmic transformation of the mpg variable

Logarithms

- $2^3 = 2 \times 2 \times 2 = 8$
- $\log_2 8 = 3$
- If $a^x = y$ then $\log_a y = x$
- Here, a is the base of the logarithm
- $\log_2 8$ asks to what power must we raise 2 to get 8
- Common bases are 10 and e (2.718282...)

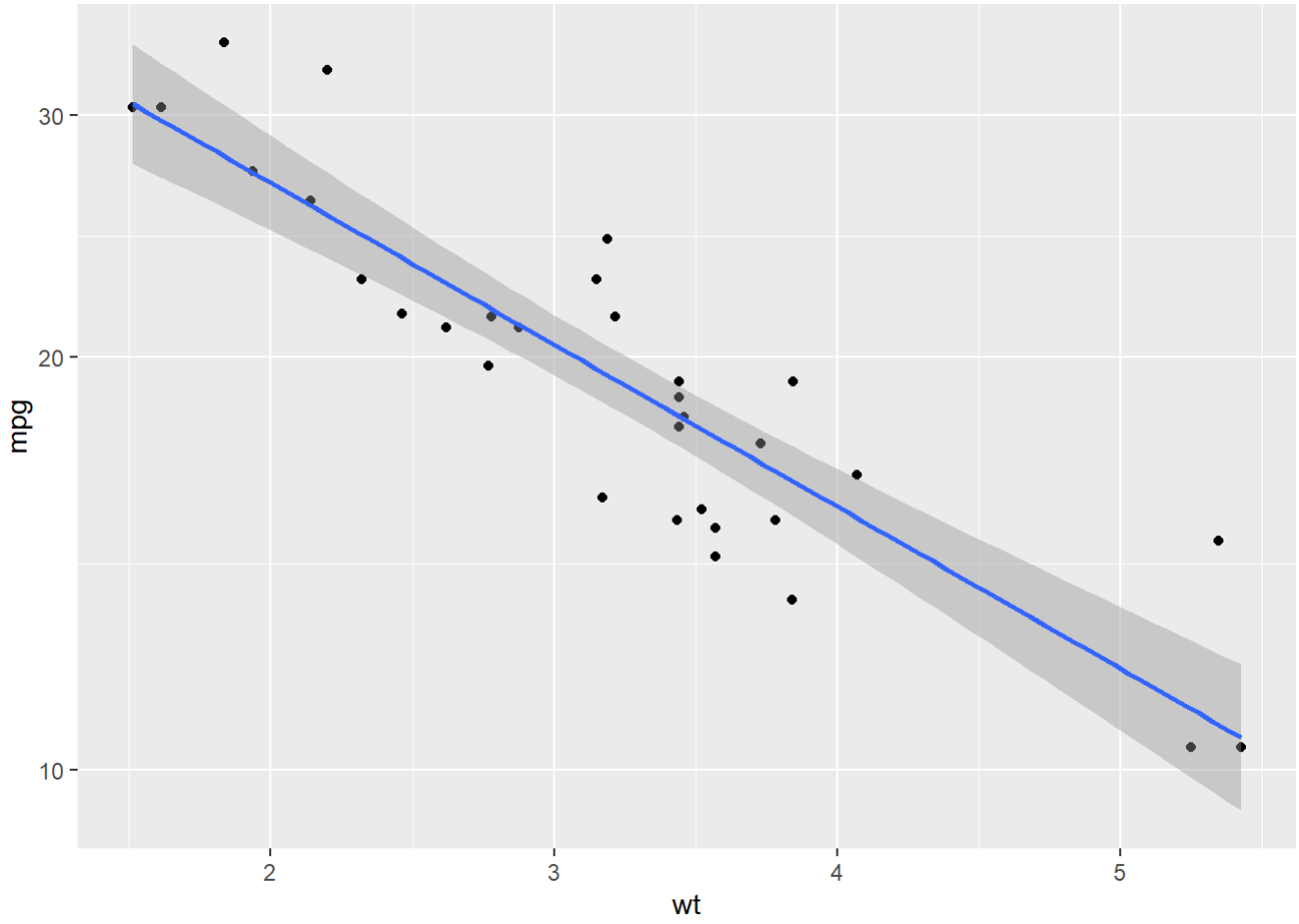
Some rules

- Logarithms are defined only for $x > 0$
- $\log(xy) = \log x + \log y$
- $\log \frac{x}{y} = \log x - \log y$
- $\log x^n = n \log x$

How does this help?

- We have estimated relationships of the form $y = \beta_0 + \beta_1 x$
- Suppose we think the relationship between x and y should be $y = \beta_0 \cdot x^{\beta_1}$
- We can't estimate this using OLS because it is nonlinear in its parameters
- Taking logs give us $\log y = \log \beta_0 + \beta_1 \cdot \log x$
- We can estimate this using OLS because it is linear in its parameters
- Let us revisit our plot but this time use a log scale for the `mpg` variable

Logarithmic transformation

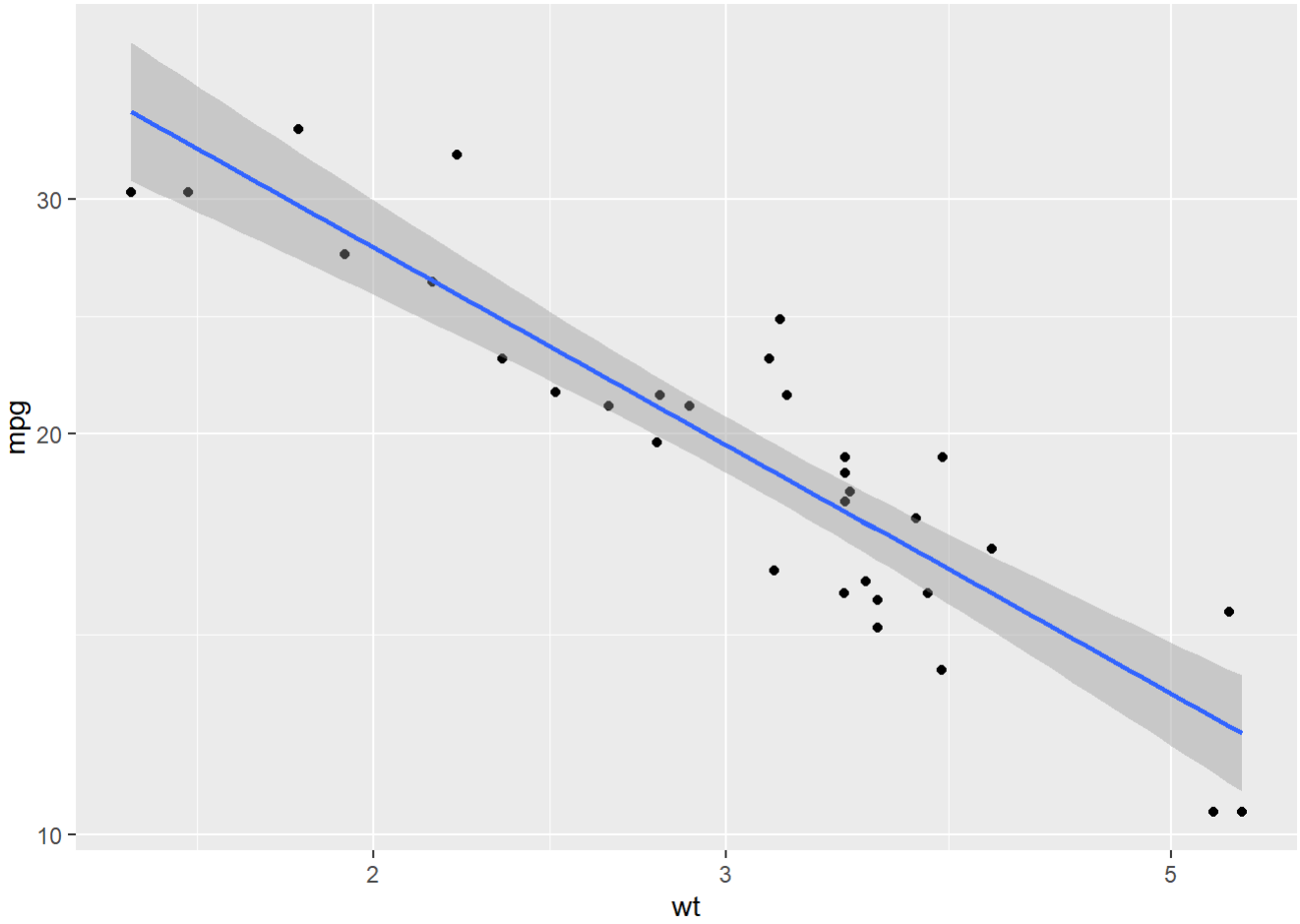


Logarithmic transformation

- This looks slightly better

- There is less of a tendency of the model to underestimate the upper and lower ends of the distribution
- We could consider also using a logarithmic transformation of the wt variable

Logarithmic transformation



Logarithmic transformation

- This looks even better
- The relationship between the variables on a log scale looks much more linear
- We can extend this logic into our regression model
- Let's consider the three corresponding models

Linear model

```
mtcars %>% lm(mpg ~ wt, data=.) %>% summary
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
## wt           -5.3445     0.5591  -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

Semi-log model

```
mtcars %>% mutate(log_mpg = log(mpg)) %>% lm(log_mpg ~ wt, data=.) %>% summary
```

```
##
## Call:
## lm(formula = log_mpg ~ wt, data = .)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.210346 -0.085932 -0.006136  0.061335  0.308623
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.83191    0.08396   45.64  < 2e-16 ***
## wt          -0.27178    0.02500  -10.87 6.31e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1362 on 30 degrees of freedom
## Multiple R-squared:  0.7976, Adjusted R-squared:  0.7908
## F-statistic: 118.2 on 1 and 30 DF,  p-value: 6.31e-12
```

Log-log model

```
mtcars %>% mutate(log_mpg = log(mpg), log_wt = log(wt)) %>% lm(log_mpg ~ log_wt, da
ta=.) %>% summary
```

```
##
## Call:
## lm(formula = log_mpg ~ log_wt, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.18141 -0.10681 -0.02125  0.08109  0.26930
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.90181    0.08790   44.39  < 2e-16 ***
## log_wt      -0.84182    0.07549  -11.15 3.41e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1334 on 30 degrees of freedom
## Multiple R-squared:  0.8056, Adjusted R-squared:  0.7992
## F-statistic: 124.4 on 1 and 30 DF,  p-value: 3.406e-12
```

Interpretation

- In the linear model, the coefficient tells us by how much the dependent variable changes when the independent variable changes by one unit
- In the semi-log model, the coefficient relates changes in the independent variable to *percentage* changes in the dependent variable
- In the log-log model, the coefficient relates *percentage* changes in the independent variable to *percentage* changes in the dependent variable

Interpretation in the semi-log model

- The coefficient multiplied by 100 ($\beta \cdot 100$) tell us *approximately* the percentage change in the dependent variable in response to a one unit increase in the independent variable
- This approximation is good when β is close to zero
- To get the exact percentage, we can calculate $100 \cdot (e^{\beta} - 1)$

Interpretation in the log-log model

- The coefficient in a log-log model tells us *approximately* the *percentage* change in the dependent variable which would result from a one *percentage* increase in the independent variable

- To get the exact value we can calculate $(1.01^{\beta} - 1) \cdot 100$
- To get the effect of a 10% increase, we can calculate $(1.1^{\beta} - 1) \cdot 100$

Our model coefficients

- In our original model, the coefficient is -5.34
- An increase in weight of one tonne is associated with a fall in the miles per gallon of 5.34
- In our semi-log model, the coefficient is -0.27
- An increase in weight of one tonne is associated with a fall in the miles per gallon of 28.03%
- In our log-log model, the coefficient is -0.84
- An increase in weight of one percent is associated with a fall in the miles per gallon of 0.83%

Selecting a model

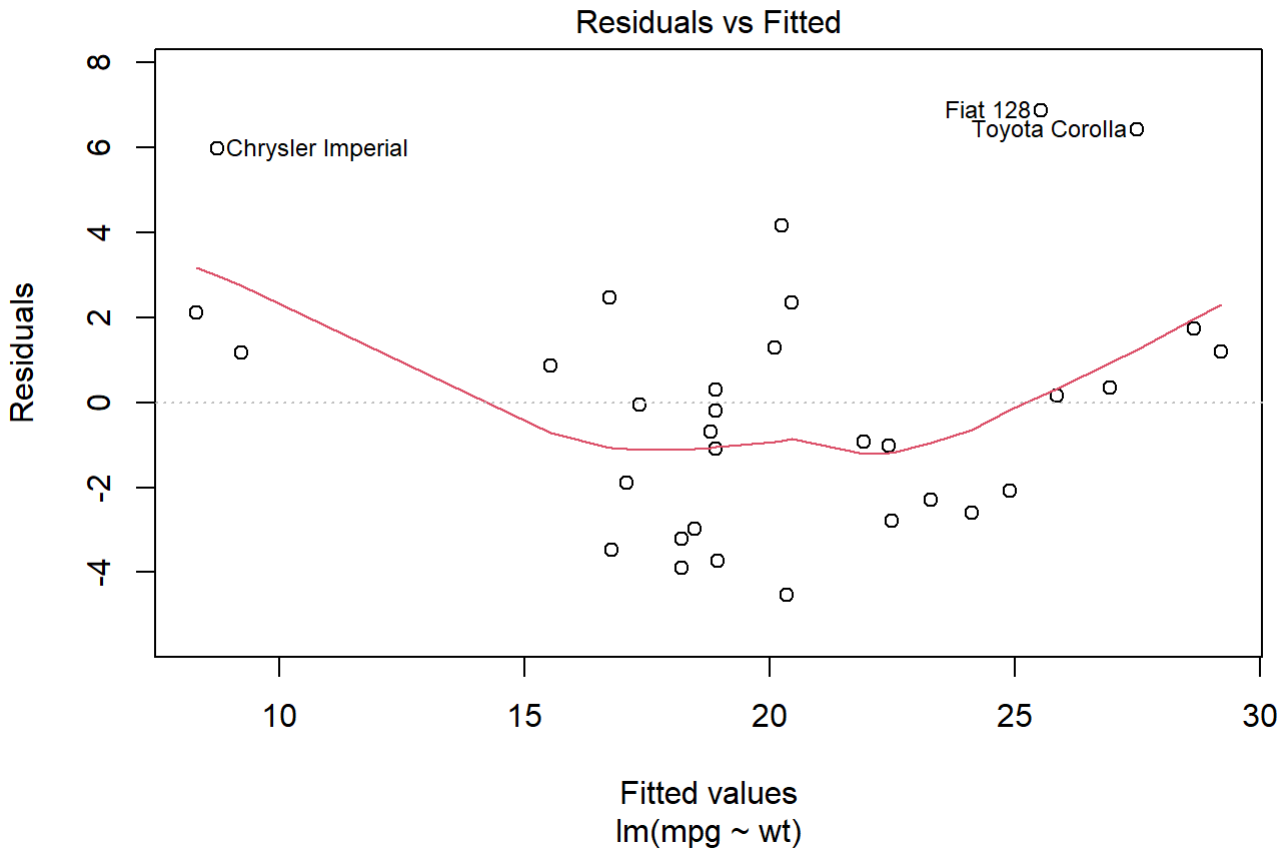
- We might have a theoretical reason *a priori* to prefer one specification over another
- Failing that we might have a preference based on our bivariate visualisation
- We could also consider the goodness of fit measures
- In this case, the log-log model had the highest R^2, although it was only marginally higher than the semi-log model
- There are also some diagnostic plots and tools we can use

The residuals versus fitted plot

- Let's return to the first regression using the mtcars dataset
- We plotted the miles per gallon against the weight
- We ran the regression $\hat{\text{mpg}}_i = \beta_0 + \beta_1 \cdot \text{wt}$
- We can get some idea of how the model performs by plotting the fitted values ($\hat{\text{mpg}}$) against the residuals
- It can help us find nonlinear patterns, as well as some other issues (more on that later)

Residuals versus fitted of our first regression

```
mtcars %>% lm(mpg ~ wt, data=.) %>% plot(which=1)
```

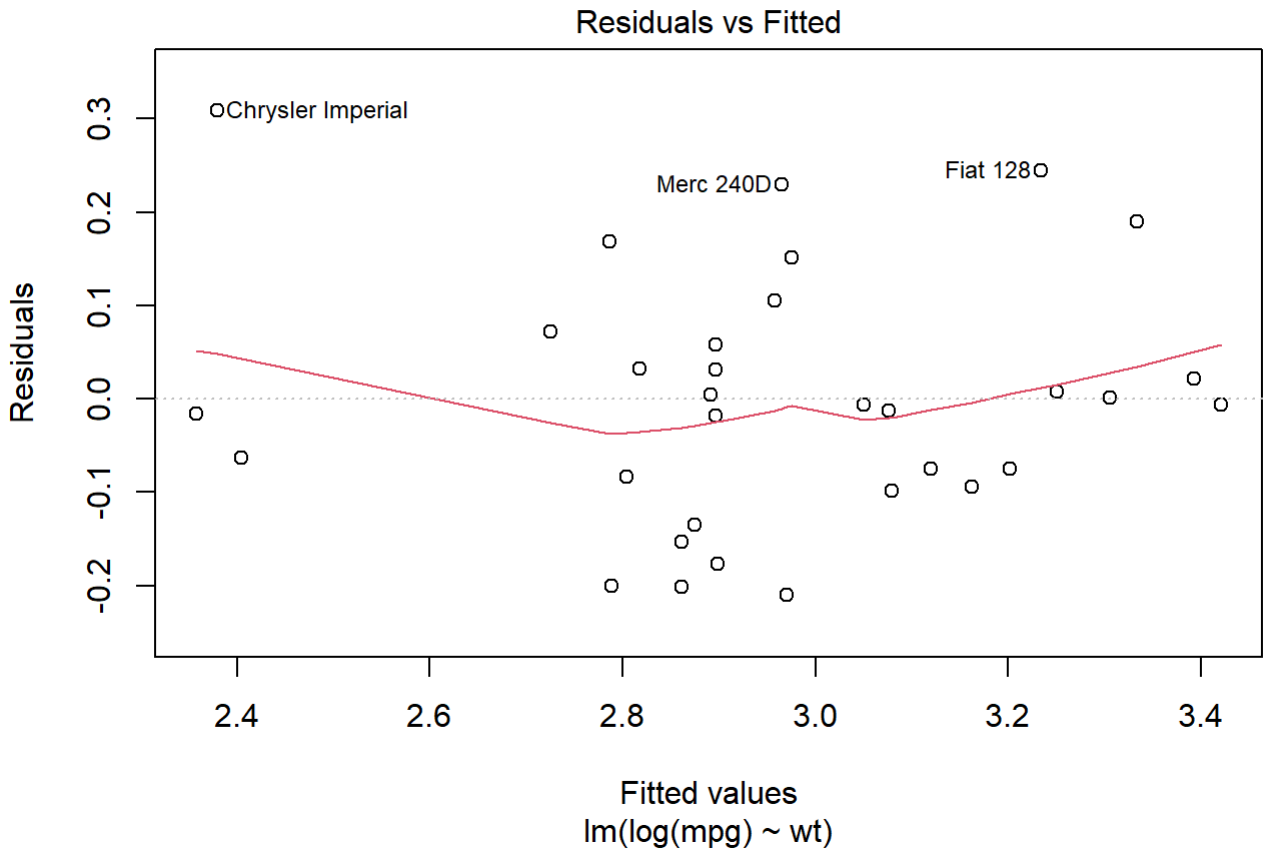


Interpretation

- This is a new way to see what we saw earlier
- The model underestimates the efficiency of the lightest and heaviest cars and overestimates the medium weight cars
- This is bad, because it means there is structure left in the residuals
- We should try to capture that structure so that all we have left in the residuals is *white noise* i.e., randomness

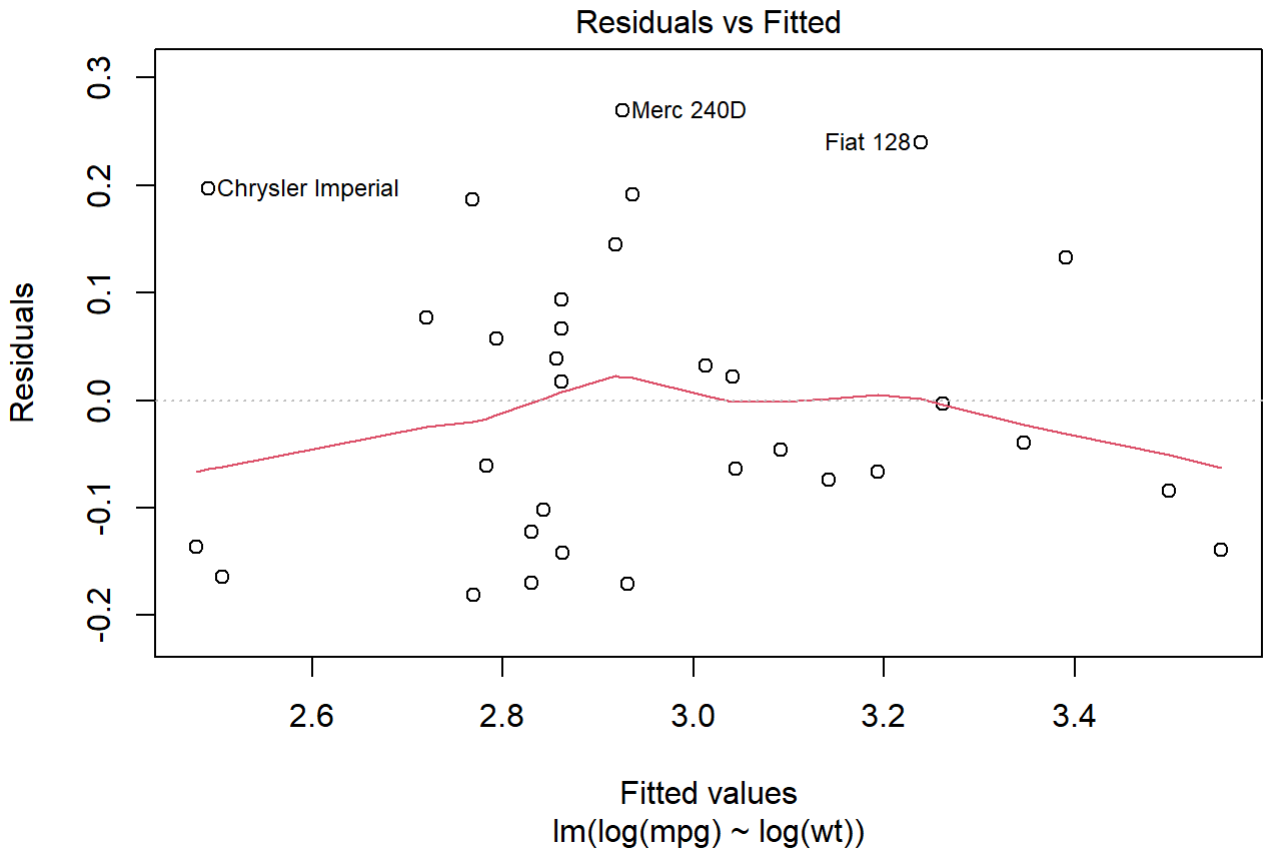
Residuals versus fitted of our second regression

```
mtcars %>% lm(log(mpg) ~ wt, data=.) %>% plot(which=1)
```



Residuals versus fitted of our third regression

```
mtcars %>% lm(log(mpg) ~ log(wt), data=.) %>% plot(which=1)
```



The RESET test

- We can use the Ramsey Regression Equation Specification Error Test (RESET) to help us decide
- The test is a general specification test which can catch a variety of types of misspecification, including nonlinearity
- If our original regression is $y_i = \beta_0 + \beta_1 \cdot x_i$, we estimate something like $y_i = \beta_0 + \beta_1 \cdot x_i + \gamma_1 \cdot \hat{y}^2 + \gamma_2 \cdot \hat{y}^3$
- We then do an F-test of the γ coefficients to see if they are statistically significant
- Rejection of the null hypothesis indicates a misspecified model
- We can use the *resettest()* function from the *lmtest* package (don't forget to install and load it)

RESET test of linear model

```
mtcars %>% lm(mpg ~ wt, data=.) %>% resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 5.1315, df1 = 2, df2 = 28, p-value = 0.01263
```

RESET test of semi-log model

```
mtcars %>% mutate(log_mpg = log(mpg)) %>% lm(log_mpg ~ wt, data=.) %>% resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 1.2329, df1 = 2, df2 = 28, p-value = 0.3068
```

RESET test of log-log model

```
mtcars %>% mutate(log_mpg = log(mpg), log_wt = log(wt)) %>% lm(log_mpg ~ log_wt, da
ta=.) %>% resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 0.75605, df1 = 2, df2 = 28, p-value = 0.4789
```

Thoughts

- In this case, it's relatively clear that the linear model wasn't great
- We saw this from:
    - The bivariate plot
    - The goodness of fit measures
    - The residuals versus fitted plots
    - The RESET test
- If we knew more about physics we would probably be able to explain why weight and fuel efficiency are nonlinear from a theoretical perspective

Back to our model

- I suggested we look at the mtcars dataset because it's a bit easier to see the nonlinear pattern there
- We can start work on our model now
- As you may have guessed it will be more complicated in our large model
- Let's start by looking at a RESET test for our model

The RESET test

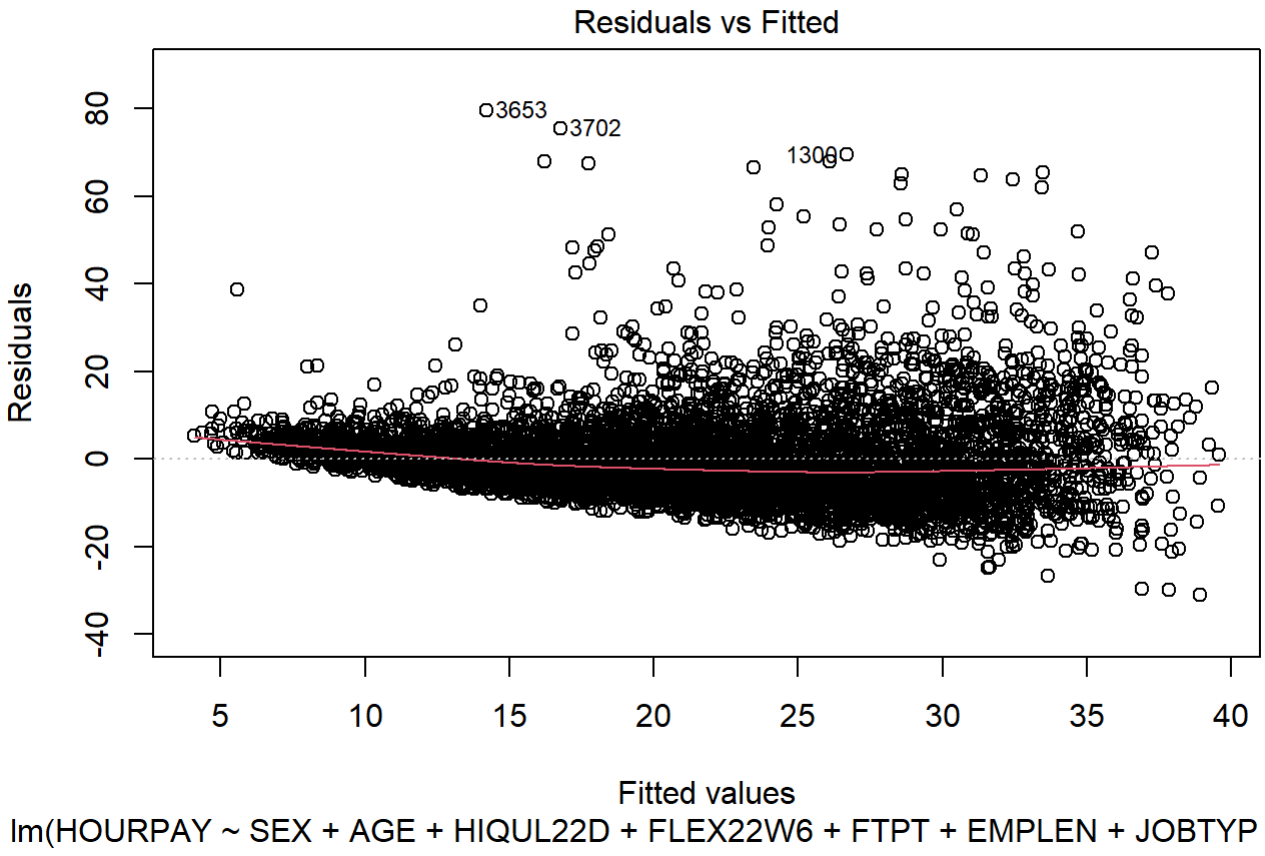
```
lfs %>%
  lm(HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 +
MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.) %>%
  resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 123.33, df1 = 2, df2 = 7289, p-value < 2.2e-16
```

RESET test

- Looks like bad news for our model
- There is a strong indication that we have misspecification of some sort
- Let's have a look at the residual versus fitted plot

Residual versus fitted



Interpretation

- The plot looks quite different to what we had before
- The red line has a bit of a curve to it
- There are also some other patterns
- There are some high-earners where we underestimate their wage
- The points seem to be more spread out for higher fitted values
- The points becoming more spread out in this way is problematic, which leads us to Assumption 2

Assumption 2

- We assume that the residuals have the same variance and are independent of one another
- Constant variance is referred to as homoscedasticity, the opposite being heteroscedasticity
- For the assumption to be satisfied, we should expect to see the points on our residuals versus fitted plot spread out evenly around the zero line
- Heteroscedasticity typically looks like a funnel shape
- There is a clear indication of this in our plot

Types of heteroscedasticity

- Heteroscedasticity can appear in our models for different reasons
- The first reason might be because we have misspecified our model, so we should always check this first
- Fixing the specification might get rid of the heteroscedasticity
- Even in a correctly specified model, we may still have heteroscedasticity, sometimes referred to as *pure heteroscedasticity*
- Pure heteroscedasticity means that our standard errors are incorrect, and therefore we cannot trust our p-values / hypothesis testing
- The problem is easily fixed though

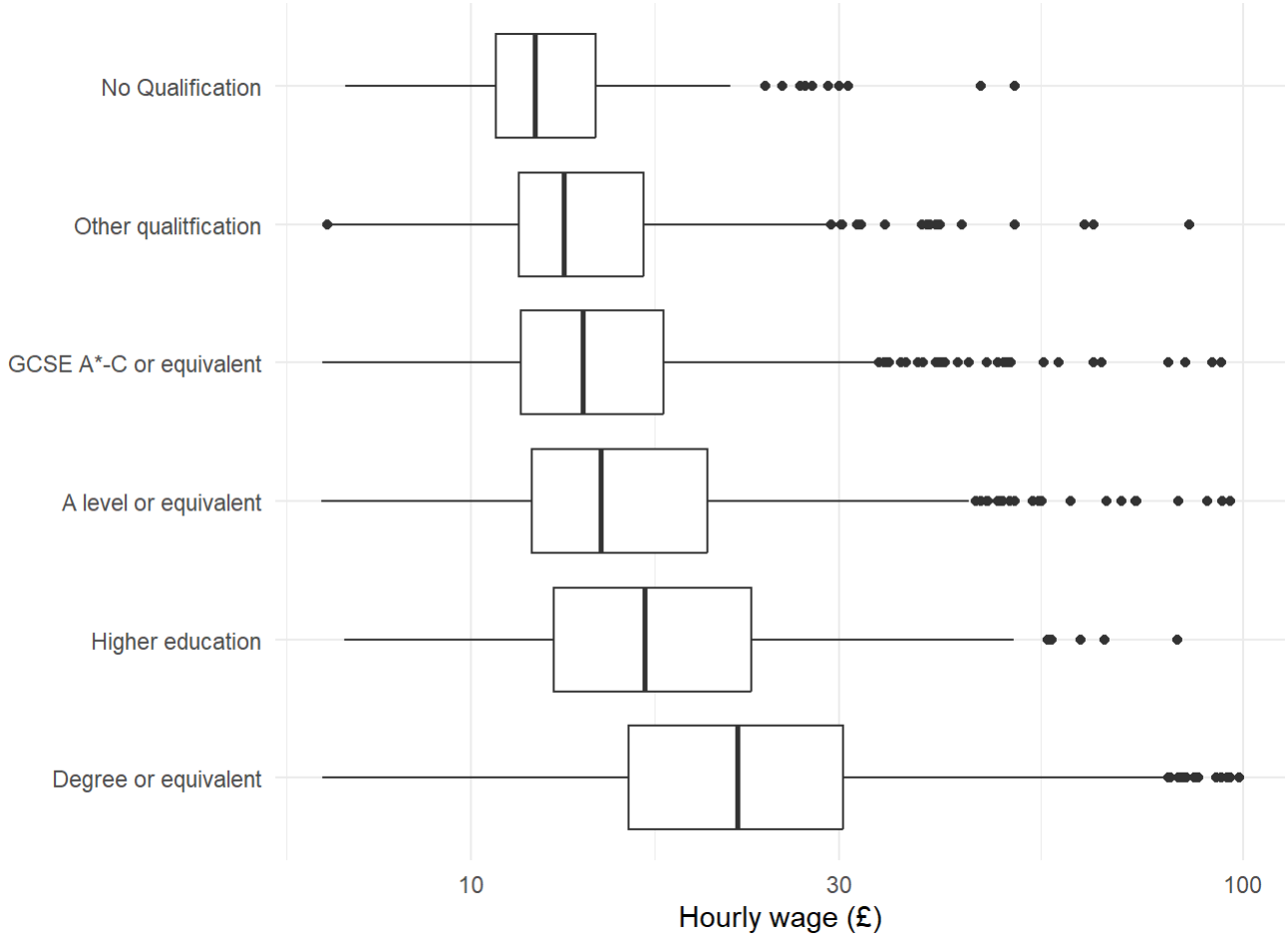
Does our model have heteroscedasticity

- First deal check that the model is specified properly
- Ask yourself whether there are any *a priori* reasons to expect heteroscedasticity
- Are there some variables which are likely to cause heteroscedasticity?
- This process starts when we design our model and when doing our descriptive statistics and data visualisation
- There are several candidates in our model e.g., education

Education

HIQUL22D	Obs	Mean	Var	StdDev
Degree or equivalent	3535	25.02	159.93	12.65
Higher education	565	19.63	97.70	9.88
A level or equivalent	1604	17.33	78.49	8.86
GCSE A*-C or equivalent	1102	16.16	68.67	8.29
Other qualitfication	331	15.63	73.18	8.55
No Qualification	210	13.53	29.07	5.39

Education



Education

- From these basic descriptions of the data, there is already an indication that education may cause heteroscedasticity
- Before even looking a the data, we should probably have expected this
- This means that even once our model is correctly specified, we will probably still end up with heteroscedasticity

Testing for heteroscedasticity

- While looking at plots and using our brain is important, it can also be helpful to use a formal test
- Several tests have been developed and these have various strengths and weaknesses
- We will look at the Breusch-Pagan test

The Breusch-Pagan test

- There are some different versions of this test
- The idea is to regress an estimate of the variance of the residuals (the square of the residual) on the independent variables

- If too much of the variation in the variance is explained, then we reject the null hypothesis of homoscedasticity
- We can use the *bptest()* function from the *lmtest* package to conduct the test
- We can run it on our model now, although there isn't much point because we have already identified an issue with the specification

Checking our model for heteroscedasticity

```
lfs %>%
  lm(HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 +
MARSTA + INDE07M + NSECMJ20 + GOR9D,
     data=.) %>%
 bptest()
```

```
##
##  studentized Breusch-Pagan test
##
## data:  .
## BP = 390.68, df = 55, p-value < 2.2e-16
```

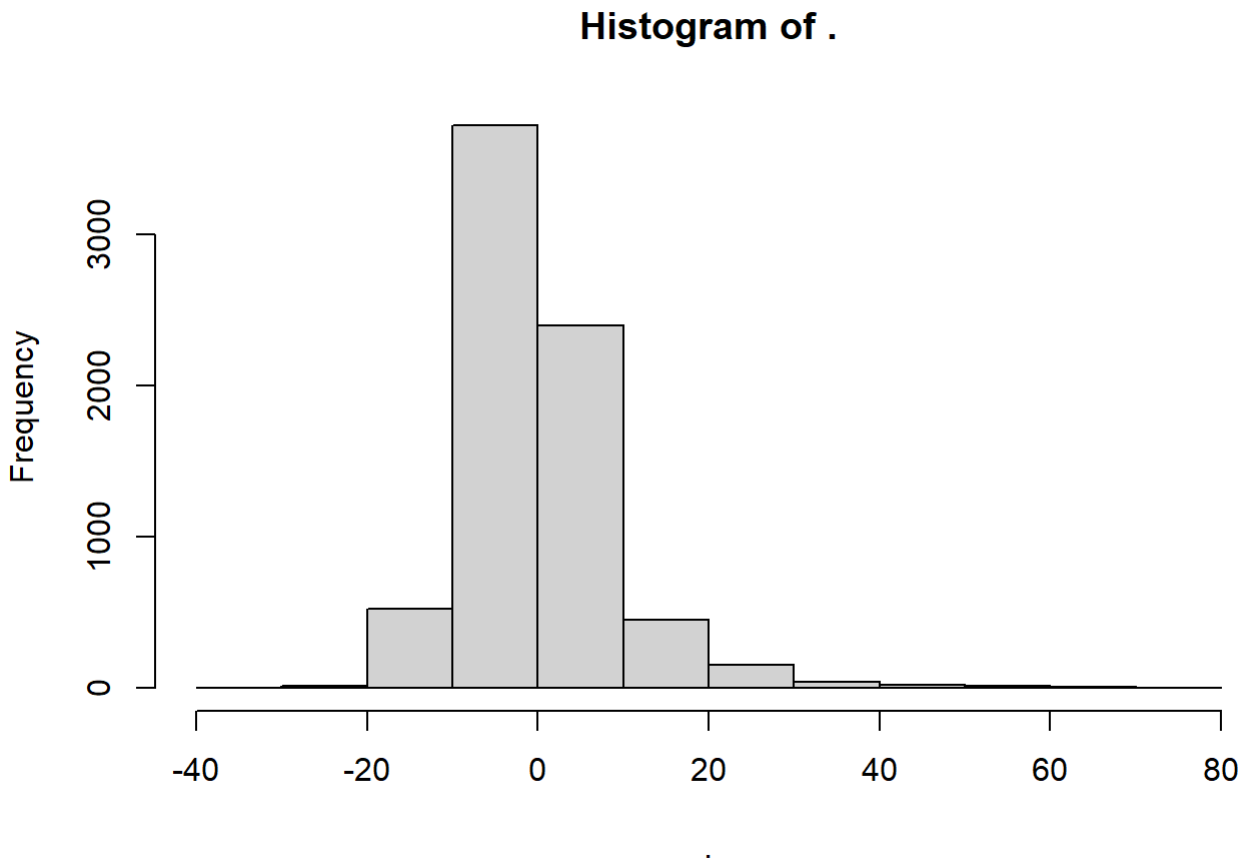
Distribution of the residuals

- We assume that the variance of the residuals is constant
- Another assumption which we make is that the follow a normal distribution
- If they do not, then we cannot reliably conduct our hypothesis testing
- There are a few ways we can examine this
- While there are statistical tests of normality, I do not recommend them as they are too sensitive
- You will almost always reject a null hypothesis of a normal distribution

Distribution of the residuals

- It is better to think about this assumption as a matter of degree
- The more normal the residuals are, the better
- We can look at a histogram of our residuals
- We can also consider a so-called QQ plot

Histogram of residuals

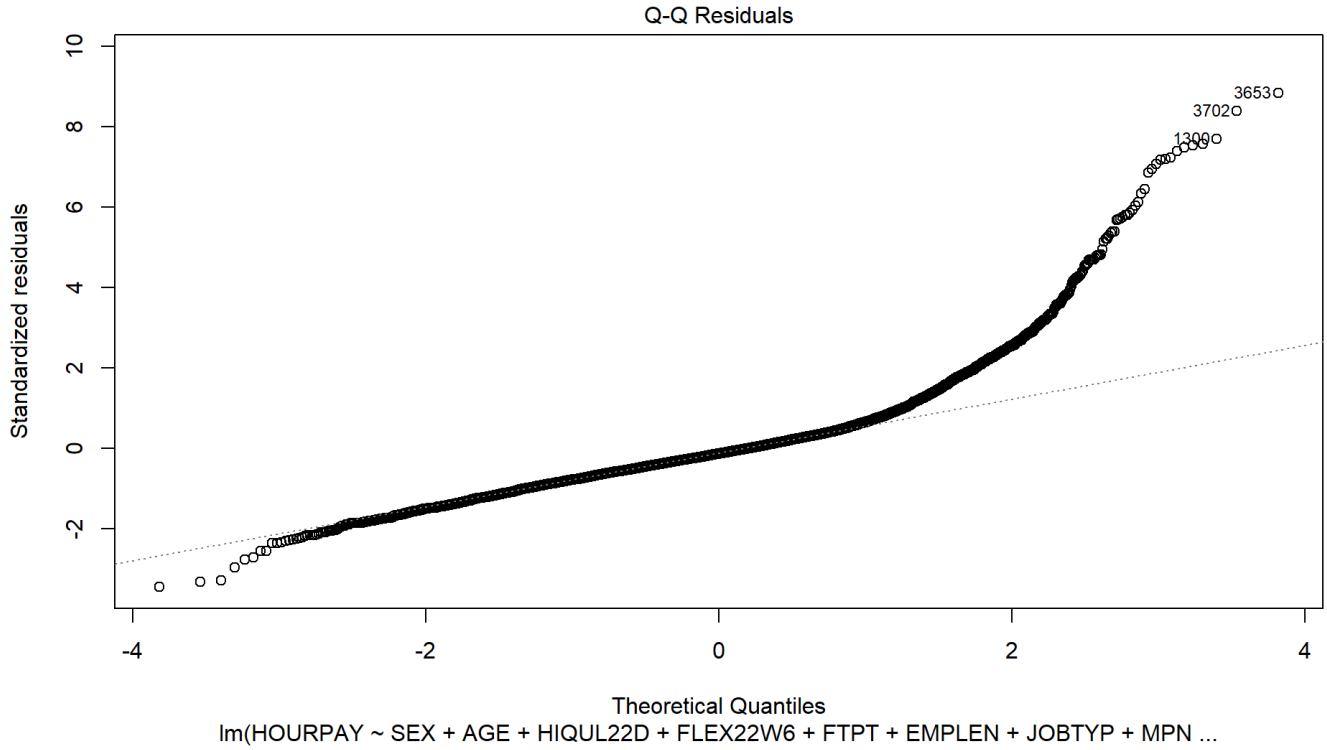


Distribution of the residuals

- The histogram doesn't look very normal
- It is highly skewed so we probably can't trust our model's p-values
- Another useful plot to look at is the quantile-quantile plot (or QQ plot)
- We are looking for the points on the chart to lie along the dotted line
- If the distribution is perfectly normal, then all the points will lie on this line

QQ plot of residuals

```
lm(HOURPAY ~ SEX + AGE +  HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + M
ARSTA + INDE07M + NSECMJ20 + GOR9D, data=lfs) %>% plot(which=2)
```



Distribution of the residuals

- The QQ plot is a different way of looking at the distribution
- Unsurprisingly, it shows the same problem i.e., our residuals are not normally distributed

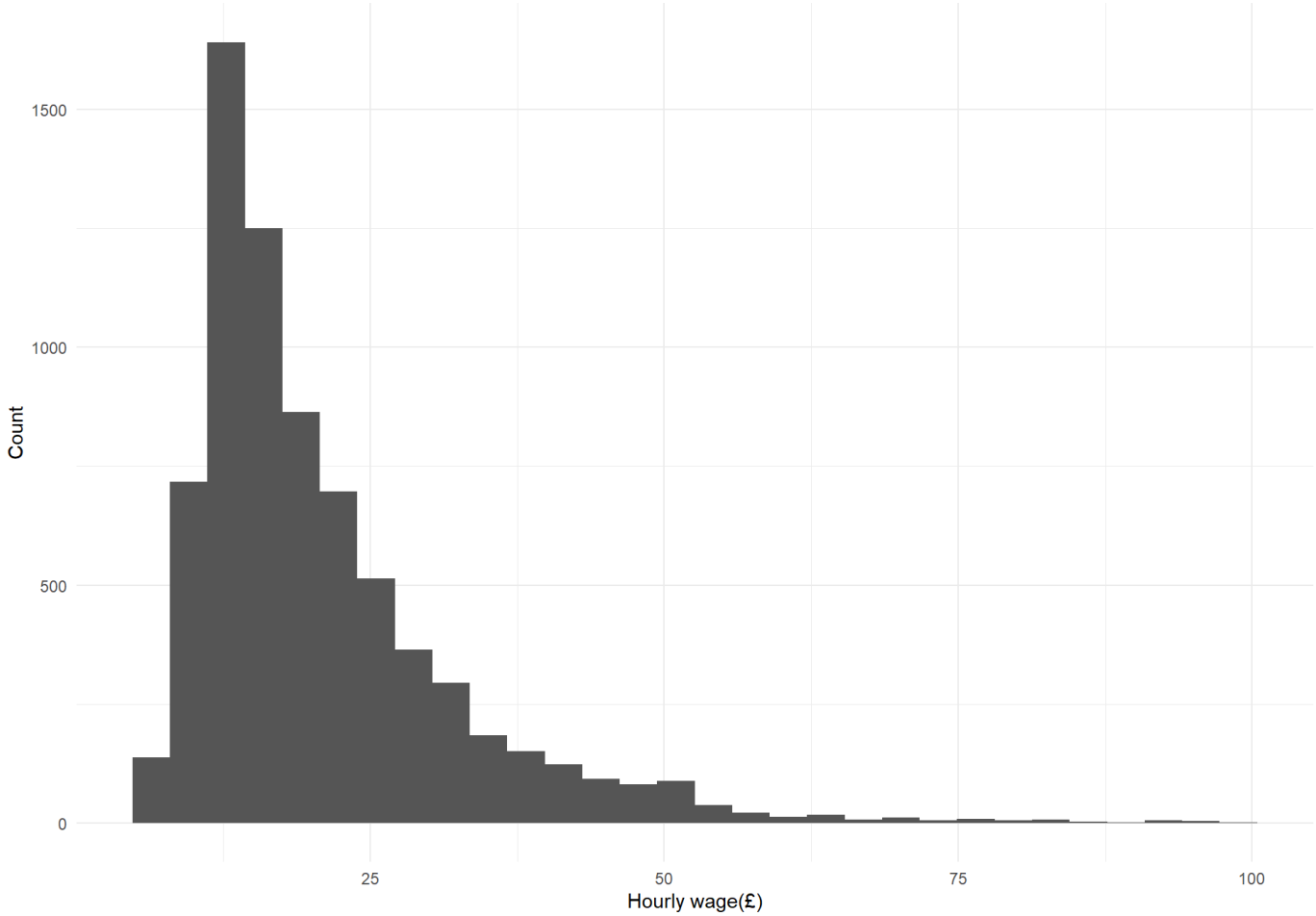
Problems

- So far, we have looked at aspects of assumptions 1 and 2
- We have an indication that our model is misspecified and that the residuals do not behave as desired
- It turns out, using the logarithmic transformation can help address all of these issues
- We will give it a go here

Distributions

- Note that the assumption of normality applies to the *residuals* and not the *variables*
- However, if our dependent variable is skewed, then our residuals will probably also be skewed
- Let's look at a histogram of our dependent variable again

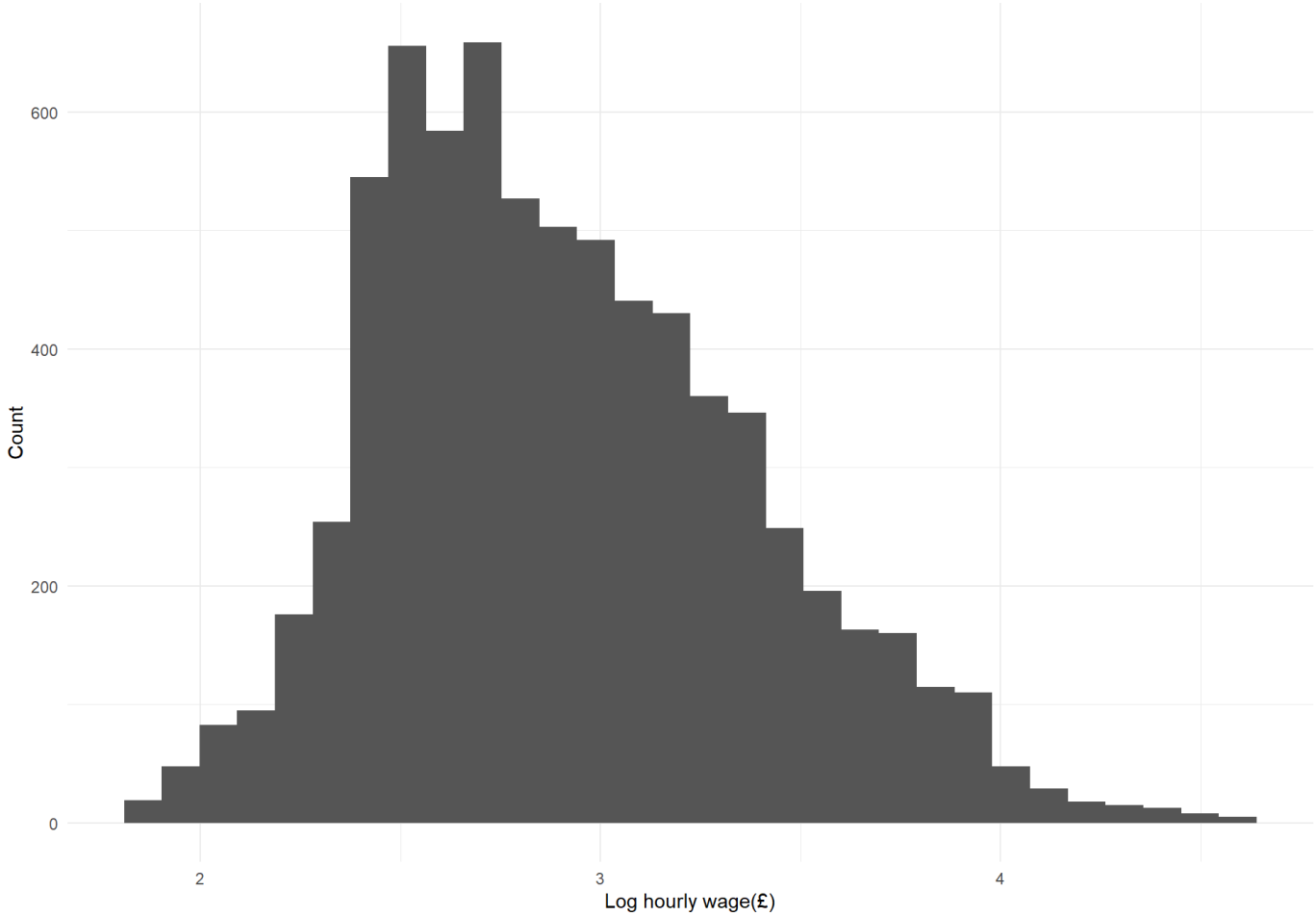
Distribution of our dependent variable



Interpretation

- We can see our variable is skewed
- In fact, it looks skewed in a similar way to our residuals
- When you see you have a dependent variable like this, you should expect to have non-normally distributed residuals
- Using a log transformation often helps

Distribution of log wages



Interpretation

- This still isn't normal
- However, it is much closer to a normal distribution that what we started with
- We can try estimating our model in a semi-log specification and see what happens

Dealing with zeros

- Note that this transformation only works if we have no zeros (as in our case)
- If you do, then a log transformation won't work because the log of zero doesn't exist
- You can consider another type of transformation, e.g., a square root transformation, but it makes interpretation of the results more difficult
- Some also suggest replacing the zero values with a small number or adding a small number to all values
- This approach is popular but may cause problems, especially if you have a large number of zeros
- Another option is to exclude the zeros and model only the positive numbers
- This limits the generalisability of the model and changes the interpretation of the results

A semi-log regression

```
lfs %>% mutate(log_wage = log(HOURPAY)) %>% lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNRO2 + MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.) %>% summary
```

```
##
## Call:
## lm(formula = log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT +
##     EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 +
##     GOR9D, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.62055 -0.21109 -0.01861  0.19648  1.96285
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                       3.0115227  0.0744636
## SEXFemale                                        -0.0719738  0.0088949
## AGE                                               0.0041173  0.0004292
## HIQUL22DHigher education                         -0.1352929  0.0158525
## HIQUL22DA level or equivalent                    -0.1692705  0.0111238
## HIQUL22DGCSE A*-C or equivalent                  -0.2011436  0.0130428
## HIQUL22DOther qualitfication                     -0.1975554  0.0209636
## HIQUL22DNo Qualification                         -0.2873597  0.0258566
## FLEX22W6No                                        0.0567135  0.0295696
## FTPTPart-time                                    -0.0297625  0.0108578
## EMPLENThree months, less than 6                  -0.0151183  0.0321268
## EMPLEN6 months, less than 12                      0.0146695  0.0295457
## EMPLEN1 year, less than 2                         0.0065315  0.0269487
## EMPLEN2 years, less than 5                        0.0477627  0.0258684
## EMPLEN5 years, less than 10                       0.0642989  0.0264364
## EMPLEN10 years, less than 20                      0.0872838  0.0265202
## EMPLEN20 years or more                            0.1164016  0.0275700
## JOBTYPNot permanent in some way                   0.0104594  0.0236075
## MPNR0211-19                                       0.0111125  0.0175732
## MPNR0220-24                                       0.0178841  0.0228220
## MPNR02Don't know but under 25                    -0.0260251  0.0296457
## MPNR0225-49                                       0.0139941  0.0154129
## MPNR0250-249                                      0.0441019  0.0132299
## MPNR02250-499                                     0.0889228  0.0187895
## MPNR02Don't know but between 50 and 499           0.0365645  0.0219740
## MPNR02500 or more                                 0.1195901  0.0134575
## MARSTAMarried, living with spouse                 0.0579014  0.0096678
## MARSTAMarried separated from spouse               0.0358534  0.0277121
## MARSTADivorced                                    0.0675272  0.0172986
## MARSTAWidowed                                     0.0260218  0.0409994
## MARSTACurrently or previously in civil partnership  0.0012794  0.0396475
## INDE07MEnergy and water                           0.1886694  0.0655100
## INDE07MManufacturing                              0.1221503  0.0610133
## INDE07MConstruction                               0.1949621  0.0623280
## INDE07MDistribution, hotels and restaurants      -0.0073309  0.0604083
## INDE07MTransport and communication                0.1702397  0.0609278
```

```
## INDE07MBanking and finance                          0.1231448  0.0602697
## INDE07MPublic admin, education and health           0.0181740  0.0599948
## INDE07MOther services                              -0.0073582  0.0626015
## NSECMJ20Lower managerial and professional          -0.2150006  0.0110268
## NSECMJ20Intermediate occupations                   -0.4471331  0.0139022
## NSECMJ20Lower supervisory and technical            -0.4203188  0.0195287
## NSECMJ20Semi-routine occupations                   -0.5387646  0.0169187
## NSECMJ20Routine occupations                        -0.5727283  0.0177126
## NSECMJ20Never worked                               -0.4505182  0.0307417
## GOR9DYorkshire and The Humber                      -0.1609020  0.0191250
## GOR9DSouth West                                    -0.1249406  0.0191410
## GOR9DWest Midlands                                 -0.1123784  0.0205305
## GOR9DSouth East                                    -0.0892086  0.0176602
## GOR9DEast of England                               -0.0759118  0.0200910
## GOR9DEast Midlands                                 -0.1569061  0.0198246
## GOR9DNorth East                                    -0.1842434  0.0247838
## GOR9DNorth West                                    -0.1410464  0.0195823
## GOR9DWales                                         -0.1764938  0.0250278
## GOR9DNorthern Ireland                              -0.1800145  0.0214698
## GOR9DScotland                                      -0.1140573  0.0199441
##                                                    t value Pr(>|t|)
## (Intercept)                                         40.443  < 2e-16 ***
## SEXFemale                                           -8.092 6.85e-16 ***
## AGE                                                  9.594  < 2e-16 ***
## HIQUL22DHigher education                            -8.534  < 2e-16 ***
## HIQUL22DA level or equivalent                      -15.217  < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                    -15.422  < 2e-16 ***
## HIQUL22DOther qualitfication                        -9.424  < 2e-16 ***
## HIQUL22DNo Qualification                           -11.114  < 2e-16 ***
## FLEX22W6No                                           1.918 0.055154 .
## FTPTPart-time                                       -2.741 0.006138 **
## EMPLENThree months, less than 6                     -0.471 0.637953
## EMPLEN6 months, less than 12                         0.497 0.619555
## EMPLEN1 year, less than 2                            0.242 0.808501
## EMPLEN2 years, less than 5                           1.846 0.064879 .
## EMPLEN5 years, less than 10                          2.432 0.015031 *
## EMPLEN10 years, less than 20                         3.291 0.001002 **
## EMPLEN20 years or more                               4.222 2.45e-05 ***
## JOBTYPNot permanent in some way                      0.443 0.657738
## MPNRO211-19                                          0.632 0.527174
## MPNRO220-24                                          0.784 0.433282
## MPNRO2Don't know but under 25                       -0.878 0.380042
## MPNRO225-49                                          0.908 0.363938
## MPNRO250-249                                         3.334 0.000862 ***
## MPNRO2250-499                                        4.733 2.26e-06 ***
## MPNRO2Don't know but between 50 and 499              1.664 0.096159 .
## MPNRO2500 or more                                    8.886  < 2e-16 ***
## MARSTAMarried, living with spouse                    5.989 2.21e-09 ***
```

```
## MARSTAMarried separated from spouse                  1.294 0.195781
## MARSTADivorced                                      3.904 9.56e-05 ***
## MARSTAWidowed                                       0.635 0.525652
## MARSTACurrently or previously in civil partnership  0.032 0.974259
## INDE07MEnergy and water                             2.880 0.003988 **
## INDE07MManufacturing                                2.002 0.045319 *
## INDE07MConstruction                                 3.128 0.001767 **
## INDE07MDistribution, hotels and restaurants        -0.121 0.903413
## INDE07MTransport and communication                  2.794 0.005218 **
## INDE07MBanking and finance                          2.043 0.041066 *
## INDE07MPublic admin, education and health           0.303 0.761955
## INDE07MOther services                              -0.118 0.906435
## NSECMJ20Lower managerial and professional         -19.498  < 2e-16 ***
## NSECMJ20Intermediate occupations                  -32.163  < 2e-16 ***
## NSECMJ20Lower supervisory and technical           -21.523  < 2e-16 ***
## NSECMJ20Semi-routine occupations                  -31.844  < 2e-16 ***
## NSECMJ20Routine occupations                       -32.334  < 2e-16 ***
## NSECMJ20Never worked                              -14.655  < 2e-16 ***
## GOR9DYorkshire and The Humber                      -8.413  < 2e-16 ***
## GOR9DSouth West                                    -6.527 7.14e-11 ***
## GOR9DWest Midlands                                 -5.474 4.55e-08 ***
## GOR9DSouth East                                    -5.051 4.49e-07 ***
## GOR9DEast of England                               -3.778 0.000159 ***
## GOR9DEast Midlands                                 -7.915 2.85e-15 ***
## GOR9DNorth East                                    -7.434 1.17e-13 ***
## GOR9DNorth West                                    -7.203 6.49e-13 ***
## GOR9DWales                                         -7.052 1.93e-12 ***
## GOR9DNorthern Ireland                              -8.385  < 2e-16 ***
## GOR9DScotland                                      -5.719 1.11e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3402 on 7291 degrees of freedom
## Multiple R-squared:  0.483,  Adjusted R-squared:  0.4791
## F-statistic: 123.9 on 55 and 7291 DF,  p-value: < 2.2e-16
```

Our original regression

```
lfs %>% lm(HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.) %>% summary
```

```
##
## Call:
## lm(formula = HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT +
##     EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 +
##     GOR9D, data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -31.140  -5.101  -1.189   3.046  79.547
##
## Coefficients:
##                                                    Estimate Std. Error t value
## (Intercept)                                        25.78853    1.98215  13.010
## SEXFemale                                          -1.78189    0.23677  -7.526
## AGE                                                 0.10231    0.01142   8.956
## HIQUL22DHigher education                           -3.23076    0.42198  -7.656
## HIQUL22DA level or equivalent                      -3.85940    0.29611 -13.034
## HIQUL22DGCSE A*-C or equivalent                    -4.53640    0.34719 -13.066
## HIQUL22DOther qualitfication                       -4.44594    0.55803  -7.967
## HIQUL22DNo Qualification                           -6.14744    0.68828  -8.932
## FLEX22W6No                                          0.40302    0.78712   0.512
## FTPTPart-time                                      -0.07943    0.28902  -0.275
## EMPLENThree months, less than 6                    -0.39861    0.85519  -0.466
## EMPLEN6 months, less than 12                       -0.18981    0.78648  -0.241
## EMPLEN1 year, less than 2                          -0.30792    0.71735  -0.429
## EMPLEN2 years, less than 5                          0.34445    0.68859   0.500
## EMPLEN5 years, less than 10                         0.52264    0.70371   0.743
## EMPLEN10 years, less than 20                        1.06104    0.70594   1.503
## EMPLEN20 years or more                              1.57930    0.73389   2.152
## JOBTYPNot permanent in some way                     0.15802    0.62841   0.251
## MPNR0211-19                                         0.01830    0.46778   0.039
## MPNR0220-24                                         0.17017    0.60750   0.280
## MPNR02Don't know but under 25                      -1.01501    0.78914  -1.286
## MPNR0225-49                                         0.05431    0.41028   0.132
## MPNR0250-249                                        0.51390    0.35217   1.459
## MPNR02250-499                                       1.50815    0.50016   3.015
## MPNR02Don't know but between 50 and 499             0.65691    0.58493   1.123
## MPNR02500 or more                                   2.40393    0.35823   6.711
## MARSTAMarried, living with spouse                   1.44619    0.25735   5.620
## MARSTAMarried separated from spouse                 0.79898    0.73767   1.083
## MARSTADivorced                                      1.68799    0.46047   3.666
## MARSTAWidowed                                       1.62508    1.09137   1.489
## MARSTACurrently or previously in civil partnership -0.38073    1.05538  -0.361
## INDE07MEnergy and water                             3.70456    1.74382   2.124
## INDE07MManufacturing                                2.19235    1.62412   1.350
## INDE07MConstruction                                 3.84059    1.65911   2.315
## INDE07MDistribution, hotels and restaurants        -0.39967    1.60802  -0.249
## INDE07MTransport and communication                  3.64347    1.62184   2.246
```

```
## INDE07MBanking and finance                          2.61233    1.60432   1.628
## INDE07MPublic admin, education and health           -0.37672    1.59701  -0.236
## INDE07MOther services                               -0.61520    1.66640  -0.369
## NSECMJ20Lower managerial and professional           -5.89892    0.29352 -20.097
## NSECMJ20Intermediate occupations                   -10.52187    0.37006 -28.433
## NSECMJ20Lower supervisory and technical             -9.81153    0.51984 -18.874
## NSECMJ20Semi-routine occupations                   -11.54158    0.45036 -25.627
## NSECMJ20Routine occupations                        -12.49647    0.47149 -26.504
## NSECMJ20Never worked                                -8.66825    0.81832 -10.593
## GOR9DYorkshire and The Humber                       -4.31942    0.50909  -8.485
## GOR9DSouth West                                     -3.69832    0.50952  -7.258
## GOR9DWest Midlands                                  -3.25102    0.54650  -5.949
## GOR9DSouth East                                     -2.42283    0.47010  -5.154
## GOR9DEast of England                                -2.23360    0.53480  -4.176
## GOR9DEast Midlands                                  -4.01808    0.52771  -7.614
## GOR9DNorth East                                     -5.24307    0.65972  -7.947
## GOR9DNorth West                                     -4.00454    0.52126  -7.682
## GOR9DWales                                          -4.97962    0.66622  -7.474
## GOR9DNorthern Ireland                               -5.08488    0.57151  -8.897
## GOR9DScotland                                       -3.28323    0.53089  -6.184
##                                                    Pr(>|t|)
## (Intercept)                                         < 2e-16 ***
## SEXFemale                                          5.87e-14 ***
## AGE                                                 < 2e-16 ***
## HIQUL22DHigher education                           2.16e-14 ***
## HIQUL22DA level or equivalent                       < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                     < 2e-16 ***
## HIQUL22DOther qualitfication                       1.87e-15 ***
## HIQUL22DNo Qualification                            < 2e-16 ***
## FLEX22W6No                                         0.608648
## FTPTPart-time                                      0.783459
## EMPLENThree months, less than 6                    0.641152
## EMPLEN6 months, less than 12                       0.809302
## EMPLEN1 year, less than 2                          0.667754
## EMPLEN2 years, less than 5                         0.616933
## EMPLEN5 years, less than 10                        0.457692
## EMPLEN10 years, less than 20                       0.132879
## EMPLEN20 years or more                             0.031434 *
## JOBTYPNot permanent in some way                    0.801461
## MPNRO211-19                                        0.968800
## MPNRO220-24                                        0.779394
## MPNRO2Don't know but under 25                      0.198407
## MPNRO225-49                                        0.894685
## MPNRO250-249                                       0.144539
## MPNRO22250-499                                     0.002576 **
## MPNRO2Don't know but between 50 and 499            0.261452
## MPNRO2500 or more                                  2.08e-11 ***
## MARSTAMarried, living with spouse                  1.98e-08 ***
```

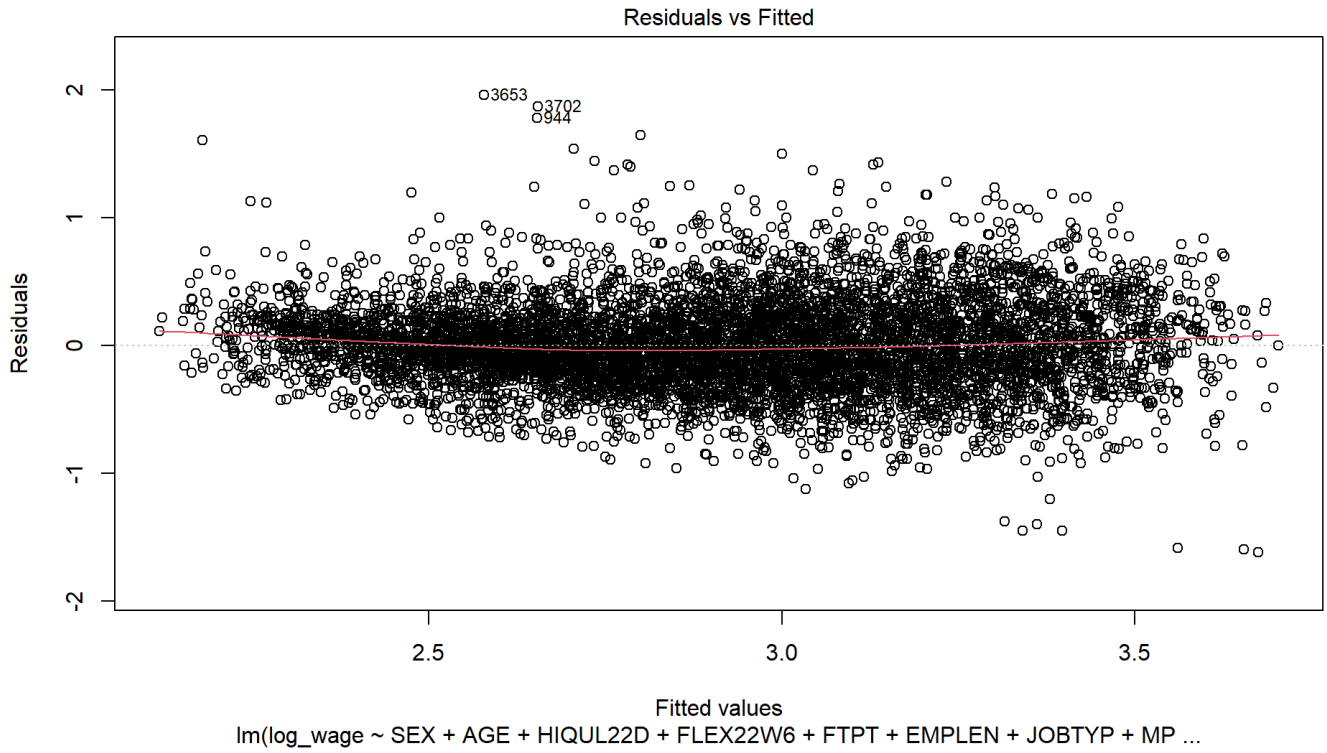
```
## MARSTAMarried separated from spouse                0.278795
## MARSTADivorced                                     0.000248 ***
## MARSTAWidowed                                      0.136521
## MARSTACurrently or previously in civil partnership 0.718297
## INDE07MEnergy and water                            0.033671 *
## INDE07MManufacturing                               0.177099
## INDE07MConstruction                                0.020649 *
## INDE07MDistribution, hotels and restaurants        0.803718
## INDE07MTransport and communication                 0.024702 *
## INDE07MBanking and finance                         0.103503
## INDE07MPublic admin, education and health          0.813522
## INDE07MOther services                              0.712006
## NSECMJ20Lower managerial and professional          < 2e-16 ***
## NSECMJ20Intermediate occupations                   < 2e-16 ***
## NSECMJ20Lower supervisory and technical            < 2e-16 ***
## NSECMJ20Semi-routine occupations                   < 2e-16 ***
## NSECMJ20Routine occupations                        < 2e-16 ***
## NSECMJ20Never worked                               < 2e-16 ***
## GOR9DYorkshire and The Humber                      < 2e-16 ***
## GOR9DSouth West                                    4.32e-13 ***
## GOR9DWest Midlands                                 2.83e-09 ***
## GOR9DSouth East                                    2.62e-07 ***
## GOR9DEast of England                               2.99e-05 ***
## GOR9DEast Midlands                                 2.99e-14 ***
## GOR9DNorth East                                    2.19e-15 ***
## GOR9DNorth West                                    1.77e-14 ***
## GOR9DWales                                         8.65e-14 ***
## GOR9DNorthern Ireland                              < 2e-16 ***
## GOR9DScotland                                      6.57e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.055 on 7291 degrees of freedom
## Multiple R-squared:  0.389,  Adjusted R-squared:  0.3844
## F-statistic: 84.41 on 55 and 7291 DF,  p-value: < 2.2e-16
```

RESET test of our semi-log model

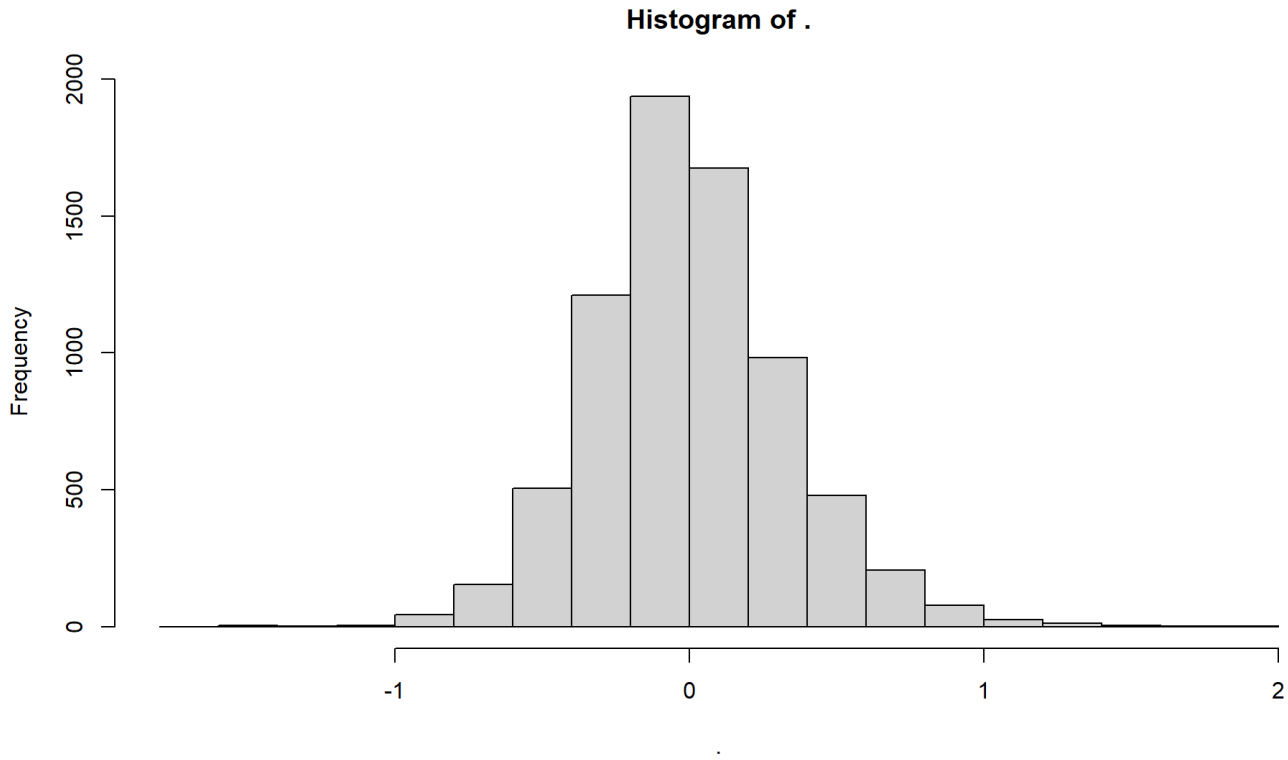
```
lfs %>% mutate(log_wage = log(HOURPAY)) %>% lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.) %>% resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 52.408, df1 = 2, df2 = 7289, p-value < 2.2e-16
```

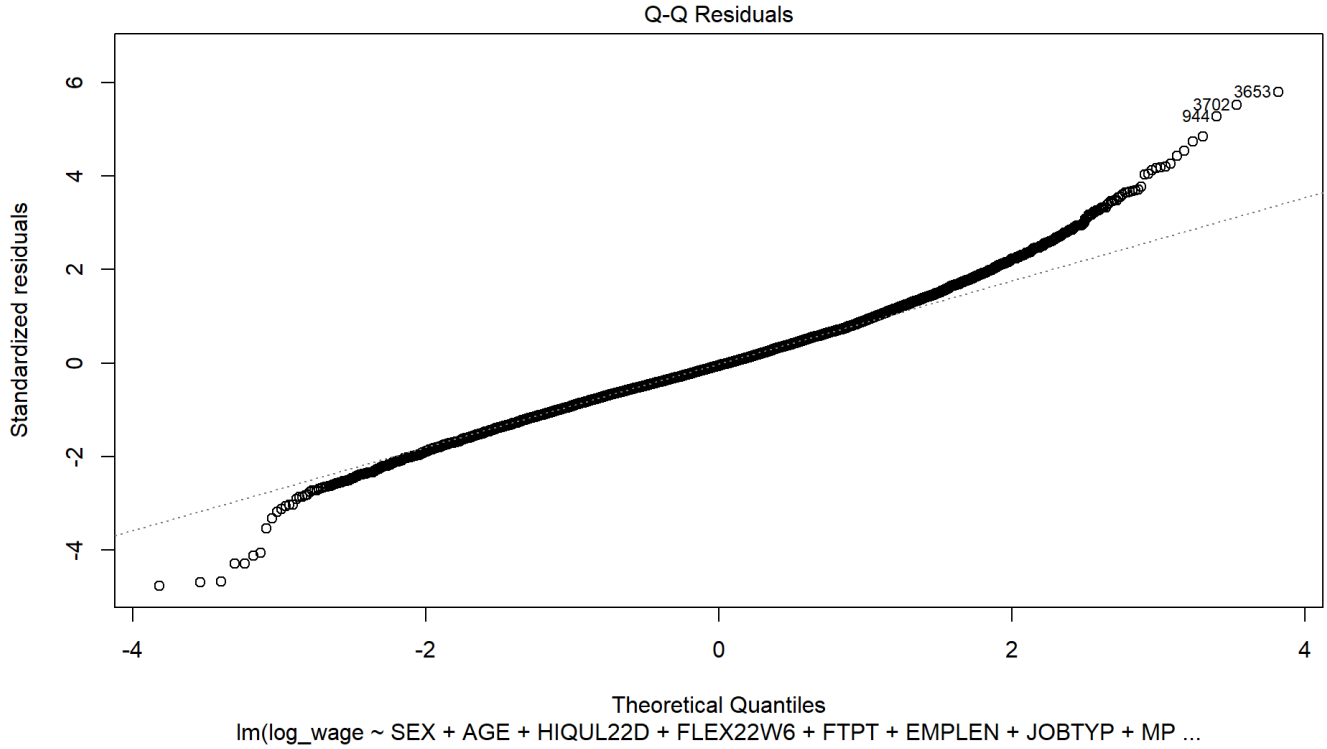
Residuals versus fitted plot of the semi-log model



Histogram of residuals from the semi-log model



QQ plot of the semi-log model



BP test of our semi-log model

```
lfs %>% mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 +
MARSTA + INDE07M + NSECMJ20 + GOR9D,
     data=.) %>%
  bptest()
```

```
##
##  studentized Breusch-Pagan test
##
## data:  .
## BP = 356.74, df = 55, p-value < 2.2e-16
```

Thoughts

- Our model still isn't passing the specification tests
- However, it is a lot better than our original model
- This is why it is important to look at both the tests and plots
- Often, we don't get our model to a perfect state
- What we are trying to do is to get it to be as good as it can be

Summary

- We have now covered some very important assumptions underlying our model
- We have looked at important diagnostic tools
- We have also started to see how we might fix some of the problems
- Next time, we will refine our model and discuss more of the assumptions